Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	8 March 2026
Team ID	Team 199
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Web Form • Registration through Email • Registration through Google Login
FR-2	User Authentication	• Login using Email and Password • Secure authentication using OTP
FR-3	Claim Data Upload	• User uploads insurance claim details • Upload customer details, claim amount, incident desc
FR-4	Data Preprocessing	• System cleans and prepares claim data • Handle missing values and normalize data
FR-5	Fraud Detection Model	• ML model analyzes claim data • Detects fraudulent or genuine caims
FR-6	Result & Report Generation	• System generates fraud prediction report • Dashboard shows Fraud / Genuine result

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system should provide a simple and user-friendly interface so that insurance staff can easily upload claim data and view fraud detection results without technical knowledge.
NFR-2	**Security**	The system must ensure secure authentication and protect sensitive insurance data using encryption and secure login me-
NFR-3	**Reliability**	The system should consistently provide accurate fraud detection results and operate without failure during claim analysis.
NFR-4	**Performance**	The system should process insurance claim data and generate fraud prediction results within a short response time.
NFR-5	**Availability**	The system should be available for users whenever required so that claim analysis and fraud detection can be performed without downtime.
NFR-6	**Scalability**	The system should be able to handle increasing amounts of insurance claim data and support more users without affecting performance.